OBJECT ORIENTED PROGRAMMING (OOP)
FOUR PILLARS OF OOP

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INTRODUCTION TO OOP

Object Oriented Programming (OOP) is a programming approach that organizes code using objects and classes.

OBJECT

An object is a real world entity that has:

- Properties (Data)
- Behaviors (Methods)

EXAMPLE

A Car

- Properties: Color, Model, Speed
- Behaviors: Start, Brake, Accelerate.

FOUR PILLARS OF OOP

Abstraction

Encapsulation

Inheritance

Polymorphism

ABSTRACTION

Abstraction means hiding unnecessary details and showing only essential features to the user.

PURPOSE OF ABSTRACTION:

- Reduces complexity
- Improves security
- Makes code easier to understand

REAL LIFE EXAMPLE:

Driving a car

- We use steering, brake, and accelerator
- We do not need to know how the engine works internally

Java

```java
abstract class Animal {
    abstract void sound();
}

class Dog extends Animal {
    void sound() {
        System.out.println("Bark");
    }
}
```

ENCAPSULATION

Encapsulation means wrapping data and methods into a single unit called a class.

PURPOSE OF ENCAPSULATION:

- Protects data
- Prevents unauthorized access
- Improves security

REAL LIFE EXAMPLE:

Bank Account

- Balance is hidden from direct access
- It can only be accessed through methods like deposit and withdraw

Java

```java
class BankAccount {
   private int balance = 5000;

   public void getBalance() {
      System.out.println(balance);
   }
}
```

INHERITANCE

Inheritance is a mechanism where one class (child class) acquires the properties and behaviors of another class (parent class).

```java
class Animal {
    void eat() {
        System.out.println("Eating");
    }
}

class Dog extends Animal {
    void bark() {
        System.out.println("Barking");
    }
}
```

PURPOSE OF INHERITANCE:

- Promotes code reusability
- Reduces code duplication
- Makes program easier to maintain

REAL LIFE EXAMPLE:

Parent to Child relationship

- Child inherits features from parents (like height, traits, etc.)

POLYMORPHISM

Polymorphism means "one thing, many forms."
It allows the same method or function to behave differently in different situations.

PURPOSE OF POLYMORPHISM:

- Increases flexibility
- Improves code readability
- Supports method reuse with different behavior

REAL LIFE EXAMPLE:

A person can act differently in different roles:

- As a student in school
- As an employee at work
- As a friend at home

POLYMORPHISM

Java

```java
class Animal {
   void sound() {
      System.out.println("Animal makes sound");
   }
}

class Dog extends Animal {
   void sound() {
      System.out.println("Dog barks");
   }
}
```

CONCLUSION

Object Oriented Programming (OOP) is a powerful programming approach that helps in building efficient and organized software systems.

THE FOUR PILLARS OF OOP ARE THE FOUNDATION OF THIS APPROACH:

- Abstraction simplifies complexity
- Encapsulation protects data
- Inheritance promotes code reuse
- Polymorphism adds flexibility

THANK YOU

We appreciate your time and attention.